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Cardiovascular Disease and Physical Activity

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In Closing



On Saturday afternoon, June 22, 2002, St. Louis Cardinals pitcher Darryl Kile was found dead in his hotel room in Chicago. The Cardinals were in Chicago to play a three-game series against the Chicago Cubs. Kile was scheduled to start the last game of the series on Sunday night. He was considered one of the Cardinals' best pitchers and a clubhouse leader. Kile, who was only 33 years old, apparently died of a heart attack caused by coronary atherosclerosis—on autopsy, two of his three major coronary arteries were found to be narrowed by 80% to 90%. Although he had no medical history or symptoms of disease, his father had died of a stroke at the age of 44 years, and Kile had complained of shoulder pain and fatigue during dinner the previous night.

On November 3, 2007, Ryan Shay, a highly ranked distance runner (National Collegiate Athletic Association 10,000 m champion in 2001 and USA marathon champion in 2003), collapsed during the U.S. Olympic marathon trials in New York City after running just 5.5 mi (8.9 km). The report on autopsy results stated that his death resulted from “cardiac arrhythmia due to cardiac hypertrophy with patchy fibrosis of undetermined etiology.” In October 2009, three men collapsed and died while running the 32nd Detroit Free Press/Flagstar Marathon, all within a span of 16 min. Several weeks earlier, two runners, one man and one woman in their mid-30s, died during the Rock 'n' Roll San Jose Half Marathon. Autopsy reports are not available for these five runners, but it is likely that their deaths were heart related since heat stress was likely not an issue. Covering the on-field collapse of a 14-year-old female soccer star who had to be revived with an AED and later underwent successful surgery for a cardiac abnormality, a 2015 CBS news report highlighted the fact that an estimated 20,000 young adults die each year from sudden cardiac arrest.

These tragedies illustrate the important fact that being a good or even an outstanding athlete during youth and young adulthood does not confer lifelong immunity to cardiovascular diseases.

Most of us consider ourselves to be healthy until we experience some overt sign of illness. With progressive diseases, such as heart disease, most people are unaware that the disease process is ongoing and could eventually cause major complications, including death. Fortunately, early detection and effective treatment of various chronic diseases can substantially reduce their severity and often avert disability and death. From a preventive standpoint, decreasing the risk factors for a disease often can either prevent the disease or

delay its onset. In this chapter, we look at cardiovascular diseases, focusing primarily on coronary artery disease (CAD) and hypertension (HTN).

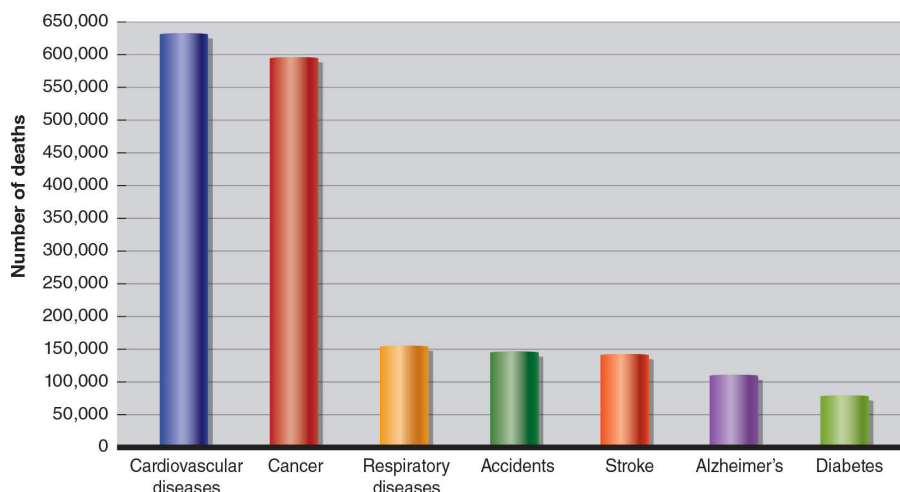


FIGURE 21.1 The leading causes of death in the United States in 2015.

Data from American Heart Association, "Heart Disease and Stroke Statistics--2014 Update," *Circulation* 129 (2014): e28-e292. <http://circ.ahajournals.org/content/129/3/e28.full>

Prevalence of Cardiovascular Disease

Heart disease is the major cause of death in the United States ([figure 21.1](#)). The following statistics have been published by the American Heart Association:⁴

- An estimated 92.1 million adults in the United States have at least one type of cardiovascular disease (CVD).
- By 2030, almost 44% of the U.S. population is projected to have some form of CVD.
- CVD was the most common cause of death worldwide, accounting for 17.3 million deaths, or 31.5% of total deaths around the world.
- Cardiovascular diseases claim more lives than all forms of cancer combined.
- The economic impact of CVD and stroke in the U.S. was estimated to be over \$316 billion in 2013-14, accounting for 14% of all health care expenditures.

But there is some good news as well:

- From 2004 to 2014, CVD death rates dropped by 25.3% and the actual number of CVD deaths decreased by 6.7%.
- Continued improvements in overall cardiovascular health metrics are projected to reduce CHD by 30% between 2010 and 2020.

From the early 1900s to the mid-1960s, the relative number of heart disease deaths, expressed per 100,000 people, increased threefold. The population of the United States more than doubled during that time, so the absolute number of heart disease deaths increased even more dramatically than the relative rate indicates. In the 1970s, cardiovascular diseases accounted for well over 50% of all deaths in the United States. They remain the number one underlying cause of death in the United States. Cardiovascular diseases continue to be a major problem in the United States, accounting for about 775,000 deaths—one in every three deaths—as of 2016.⁴

As noted previously, CVD death rates dropped by about 25% over the decade from 2004 to 2014. The reasons for this decline have been heavily debated but likely include a greater focus on disease prevention. These are some examples:

- Improved public awareness of risk factors and symptoms
- Increased use of preventive measures, including lifestyle changes (e.g., nutrition, exercise, stress reduction, and smoking cessation) to reduce individual risk
- Earlier and more accurate diagnosis
- Greater awareness and use of cardiopulmonary resuscitation techniques

Another probable reason is better treatment of patients with disease, including these examples:

- Improved drugs for specific treatment
- Angioplasty, drug-coated stents, and bypass surgery
- Greater focus on secondary prevention

Cardiovascular disease incidence and mortality rates vary by sex, age, and race. [Table 21.1](#) shows rates of all-cause deaths and death

from cardiovascular disease in the United States by sex and race, adjusted for age.

Forms of Cardiovascular Disease

There are several different cardiovascular diseases. In this section, we focus primarily on those that are preventable and that affect the largest number of individuals each year. According to the World Health Organization, an estimated 17.7 million people died from CVD in 2015. Of these deaths, an estimated 7.4 million were due to coronary artery disease and 6.7 million were due to stroke.⁴³ Other types of cardiovascular diseases include congenital diseases of the heart, cardiomyopathy, valvular heart disease, venous thromboembolism, and peripheral artery disease.

TABLE 21.1 U.S. Death Rates per 100,000 Adults by Sex and Race

Category	All cause	Cardiovascular disease
All people	733	169
Men	863	212
White	862	211
Black or African American	1,040	258
Hispanic or Latino	629	146
Women	624	134
White	627	132
Black or African American	711	166
Hispanic or Latina	438	93

These data are age standardized (adjusted to account for younger or older average age of a population).

Data from World Health Organization, *Global Atlas on Cardiovascular Disease Prevention and Control*, edited by S. Mendis, P. Puska, and Bo Norrving (World Health Organization, Geneva, 2011). Available http://whqlibdoc.who.int/publications/2011/9789241564373_eng.pdf?ua=1

Coronary Artery Disease

As most humans age, their coronary arteries (see [figure 6.4](#)), which supply blood to the myocardium (heart muscle), become progressively narrowed as a result of the formation of fatty **plaque** under the inner lining (the endothelium) of the artery, as seen in [figure 21.2](#). This progressive narrowing of the arteries in general is referred to as atherosclerosis, and when the coronary arteries are involved, it is termed **coronary artery disease (CAD)**, or coronary heart disease. As the disease progresses and the coronary arteries become narrower, the capacity to supply blood to the myocardium is progressively reduced.

As the narrowing worsens, there is a mismatch between oxygen supply to the myocardium (heart muscle) and its demand, which increases with exercise. When this occurs, the area of the myocardium that is supplied by the narrowed artery or arteries becomes ischemic, meaning that it suffers a deficiency of blood (with oxygen and nutrients). **Ischemia** of the myocardium often causes mild to severe discomfort, referred to as *angina pectoris* or simply *angina*. Anginal symptoms are often first experienced during physical exertion or stress, when the oxygen demands of the heart increase, creating a mismatch between supply and demand.

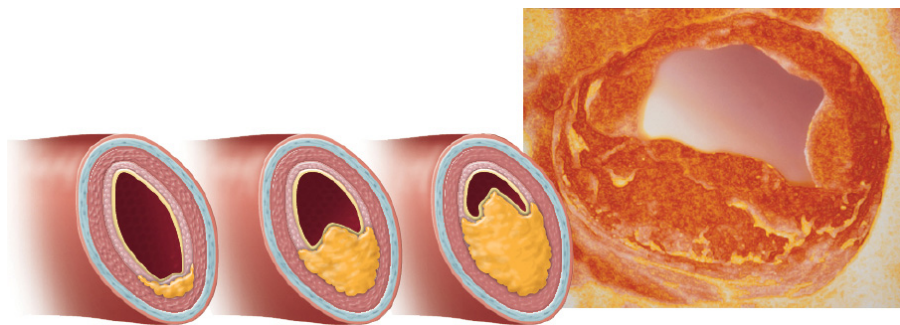


FIGURE 21.2 The progressive formation of plaque in a coronary artery.

When blood supply to a part of the myocardium is severely or totally restricted, ischemia can lead to a heart attack, or **myocardial infarction**, as cardiac muscle cells that are deprived of adequate blood flow (and thus deprived of oxygen) for several minutes undergo damage (injury) and sometimes necrosis (cellular death). The outcome of this event depends on the size of the affected myocardial area and the extent of the damage. Ischemic injury is reversible, while infarcted tissue is permanently damaged. Sometimes a heart attack is so mild that the victim is unaware that it has occurred. In such cases, the heart attack may only be discovered weeks, months, or even years later when an electrocardiogram is obtained during a routine medical examination.

Atherosclerosis is a progressive disease process, and the pathological changes that lead to atherosclerosis can begin in infancy and progress during childhood.²⁵ **Fatty streaks**, or lipid deposits, which are thought to be the precursors of atherosclerosis, are commonly found in the aortas of children by age 3 to 5. These

fatty streaks start to appear in the coronary arteries during the early teens, can develop into fibrous plaques during one's 20s, and can progress to unstable or complicated lesions during one's 40s and 50s.

RESEARCH PERSPECTIVE 21.1

Long-Term Marathon Running Reduces Coronary Artery Plaque Formation in Women

Regular endurance exercise leads to improvements in cardiovascular health and reduces cardiovascular disease risk factors, such as high blood pressure and high blood cholesterol. Typically, marathon training causes a large increase in cardiorespiratory fitness and a reduction in cardiovascular risk profiles. However, fairly little is known about the effects of long-term marathon participation on the coronary vasculature. Two studies of middle-aged men who were long-term marathon runners (classified in one study as having run more than five races in the past 3 years, and in the other study as having completed one or more marathons a year for 25 or more consecutive years) found that these men paradoxically had *greater* accumulations of plaque on their coronary arteries compared to sedentary age-matched men. Although these findings were surprising, until recently, no study had examined this phenomenon in women.

A recent study conducted at the Abbott Northwestern Hospital in Minneapolis, Minnesota, examined coronary artery plaque formation and the prevalence of cardiovascular disease risk factors in women who were long-term marathon runners.³⁶ In order to do this, researchers recruited 26 women who had run the Twin Cities Marathon for at least 10 consecutive years and 28 control women who were sedentary. Using coronary computed tomography angiography (CCTA), the researchers identified any plaque formation in the coronary arteries. At the time of the CCTA scan, participants also had a 12-lead ECG, resting heart rate and blood pressure measurements, and a blood draw for the measurement of serum lipids and creatinine. Five of the twenty-six marathon runners and fourteen of the twenty-eight controls had calcified plaques in their coronary arteries, a statistically lower incidence in the marathon running group. The five female runners with coronary plaque had run marathons for more years but were on average 12 years older than the runners without plaque. The marathon runners also had lower resting heart rate, body weight, blood pressure, and serum triglycerides and higher high-density lipoproteins compared to controls. Women who are long-term marathon runners are less likely to have coronary artery plaque formation and have a reduced cardiovascular disease risk profile compared to sedentary age-matched controls. Instead, the

development of coronary plaque in women marathoners appears related to older age and increased cardiac risk.

The rate at which atherosclerosis progresses is determined largely by genetics and lifestyle factors, including smoking history, diet, and physical activity. For some people, the disease progresses rapidly, with a heart attack occurring at a relatively young age—in their 20s or 30s. For others, the disease progresses very slowly, with only minor evidence of atherosclerosis even late in life. Most people fall somewhere between these two extremes.

To illustrate this, a study of combat fatalities from the Korean War revealed that a large percentage of autopsied American soldiers, with an average age of 22 years, already had evidence of coronary atherosclerosis.¹⁵ The extent of disease ranged from fibrous thickening to complete occlusion of one or more of the main branches of the coronary arteries. Evidence of coronary atherosclerosis also was found in almost half of the American fatalities from the Vietnam War, some with severe manifestations of the disease.³⁰

Hypertension

Hypertension is the medical term for high blood pressure, a condition in which blood pressure is chronically elevated above levels considered desirable or healthy. Blood pressure depends in part on body size, so children and young adolescents have much lower blood pressures than adults. For this reason, defining what constitutes HTN in the growing child and adolescent is difficult. Clinically, HTN in these groups is defined as blood pressure values above the 90th or the 95th percentile for the youth's age. Hypertension is uncommon during childhood but can appear during mid-adolescence.

Historically, normal blood pressure and stages of HTN were classified by well-established blood pressure cutoffs. For example, prehypertension was defined as systolic blood pressure (SBP) between 120 and 139 mmHg or a diastolic blood pressure (DBP) between 80 and 89 mmHg and stage 1 HTN was defined as SBP of 140 to 159 or DBP of 90 to 99 mmHg. The Joint National Committee

on Detection, Evaluation, and Treatment of High Blood Pressure has recently published new evidence-based guidelines (JNC8) for determining and treating HTN in adults.²⁴ The JNC8 guidelines eliminate categories of HTN (normal, pre-HTN, stage 1, and stage 2) based on blood pressure cutoffs and simply recommend treatment for specific population groups.

New treatment goals are summarized in [table 21.2](#).

These new recommendations also strongly endorse lifestyle modification guidelines, with diet and increased physical activity at the core of therapeutic management of HTN. Other major changes include altering the classes of medications recommended for the management of HTN and new recommendations for treating persons with chronic kidney disease. In hypertensive African Americans, for example, the guidelines recommend the use of thiazide-type diuretics or calcium channel blockers because these classes of drugs have higher efficacy rates in this population than other antihypertensive drugs, such as angiotensin receptor blockers or angiotensin-converting enzyme blockers.

The JNC8 panel's attempt to simplify the guidelines for treating HTN received some criticism but has generally been embraced by health care professionals. Perhaps the most significant and interesting point to be taken from JNC8 is that, unlike previous guidelines, these are evidence based, and the reason for the significant changes is the lack of evidence for some previously published treatment standards.

Hypertension causes the heart to work harder than normal because it has to pump blood from the left ventricle against a greater resistance (afterload). Furthermore, HTN places increased chronic strain on the systemic arteries and arterioles. Over time, the stress can cause the heart to enlarge and the arteries and arterioles to become less elastic. Eventually, this can lead to atherosclerosis, heart attacks, heart failure, stroke, and kidney failure.

TABLE 21.2 JNC8 Guidelines for the Management of Hypertension in Adults

Population group	Treatment goal
Age ≥60 years with SBP ≥150 or DBP ≥90	SBP <150 and DBP <90

General population <60 years with SBP >140 or DBP >90

Diabetes ≥18 years

SBP <140
DBP <90
SBP <140
DBP <90

Note. All blood pressures are in mmHg. SBP = systolic blood pressure; DBP = diastolic blood pressure.

In 2014, almost 86 million American adults, or about 34% of the adult population, were estimated to have high blood pressure (as defined by systolic ≥ 140 mmHg, diastolic > 90 mmHg, or both).⁴ Among Americans age 60 and up, over 65% have high blood pressure. Like CAD, it is most prevalent (42%) among non-Hispanic black adults.²⁴ Compared with white Americans, black Americans develop high blood pressure at an earlier age, and it is more severe at any decade of life. Consequently, black Americans have a 1.3 times greater rate of nonfatal stroke, a 1.8 times greater rate of fatal stroke, a 1.5 times greater rate of heart disease deaths, and a 4.2 times greater rate of end-stage renal disease when compared with white Americans.³ Sadly, of those diagnosed with high blood pressure, only about 50% have controlled HTN. This means that large numbers of Americans who are being seen by a physician for high blood pressure remain hypertensive.

Stroke

Stroke is a form of cardiovascular disease that affects the cerebral arteries, those that supply the brain. Globally, stroke was the cause of about 6.5 million deaths in 2013.⁴ As with other forms of CVD, the death rate from stroke have decreased significantly in recent years.

Strokes generally fall into two categories: **ischemic stroke** and **hemorrhagic stroke**. Ischemic strokes are the most common (~87% of all cases) and result from an obstruction within a cerebral blood vessel that limits the flow of blood to that region of the brain. Obstructions have one of two causes:

- Cerebral thrombosis, the most common, in which a thrombus (blood clot) forms in a cerebral vessel, often at the site of atherosclerotic damage to the vessel. This thrombus is very similar in etiology to the blood clots formed in coronary arteries that cause myocardial ischemia, injury, and infarction.

- Cerebral embolism, in which an embolus (an undissolved mass of material, such as fat globules, or bits of tissue, a blood clot) breaks loose from another site in the body and lodges in a cerebral artery. An irregular heartbeat, atrial fibrillation, creates conditions in which clots can form in the heart, dislodge, and then travel to the brain.

In cases of ischemic stroke, blood flow distal to the blockage is restricted, and the part of the brain that relies on that supply becomes ischemic—oxygen deficient—and can die. Ischemic stroke is caused by the same vascular disease process present in atherosclerosis (see earlier) and other obstructive vascular diseases, including peripheral vascular disease.

Hemorrhagic strokes are of two major types:

- Intracerebral hemorrhage, in which one of the cerebral arteries ruptures in the brain
- Subarachnoid hemorrhage, in which one of the brain's surface vessels ruptures, dumping blood into the space between the brain and the skull

In both cases, blood flow beyond the rupture is diminished because as blood leaks from the vessel at the site of injury. Because the brain is enclosed in the rigid skull, as the blood accumulates outside of the vessel, it puts pressure on the fragile brain tissue, which can alter brain function. Brain hemorrhages often result from aneurysms, weak spots in the vessel wall that balloon outward, and aneurysms may arise because of HTN or atherosclerotic damage to the vessel wall. Arteriovenous malformations, clusters of abnormally formed blood vessels, are another cause of hemorrhagic strokes.

As with a heart attack, stroke can result in death of the affected brain tissue. The consequences depend largely on the location and extent of the stroke. Brain damage from a stroke can affect the senses, speech, body movement, thought patterns, and memory. Paralysis on one side of the body is common, as is the inability to verbalize thoughts. Most effects of a stroke are indicative of the side of the brain that was damaged. One side of the brain controls the functions of the opposite side of the body. For example, a stroke on the right side of the brain may have the following effects:

- Paralysis on the left side of the body
- Vision problems
- Quick, inquisitive behavioral style
- Memory loss

Alternatively, a stroke on the left side of the brain more often results in the following:

- Paralysis on the right side of the body
- Speech and language problems
- Slow, cautious behavioral style
- Memory loss

Heart Failure

Heart failure is a chronic and progressive clinical condition in which the heart muscle (myocardium) becomes too weak to maintain a cardiac output adequate to meet the body's blood and oxygen demands. When cardiac output is inadequate, blood may pool in the veins. This causes excess fluids to accumulate in the body, particularly in the legs and ankles. This fluid accumulation (edema) also can affect the lungs (pulmonary edema), disrupting breathing and causing shortness of breath. For that reason, this condition is commonly called *congestive heart failure (CHF)*. It usually results from damage to the myocardium, which may have been caused by myocardial infarction, severe HTN, valvular heart disease, or cardiomyopathy.

Other Cardiovascular Diseases

Other cardiovascular diseases include peripheral vascular diseases, valvular heart diseases, and congenital heart diseases.

Peripheral vascular disease involves the systemic arteries and veins, as opposed to the coronary vessels. Peripheral artery disease (PAD) is atherosclerotic disease most often affecting arteries supplying blood to the legs, brain, or, more rarely, other organs. PAD in the legs can result in severe restrictions in activity due to pain. In arteries feeding the brain, PAD can result in *transient ischemic attacks (TIAs)*; stroke-like symptoms caused by temporary restriction in blood flow to the brain).

Peripheral venous diseases include varicose veins and phlebitis. Varicose veins result from incompetency of the valves in the veins, allowing blood to back up in the veins and causing them to become enlarged, tortuous, and painful. Phlebitis is inflammation of a vein and is also very painful.

Valvular heart diseases involve one or more of the four valves that control the direction of blood flow into and out of the four heart chambers. **Rheumatic heart disease** is one form of valvular heart disease resulting from rheumatic fever, a streptococcal infection that typically occurs in children between age 5 and 15. Rheumatic fever is an inflammatory disease of the connective tissue and commonly affects the heart, specifically the heart valves. The damage to the valves can cause narrowing of the opening (*stenosis*); difficulty in valves opening, hindering blood flow out of the pre-ventricular chamber; or incomplete valve closure (*prolapse* or *regurgitation*), allowing blood to leak back into the pre-ventricular chamber. The disease is typically referred to by the valve affected and the type of problem, e.g., mitral stenosis.

Congenital heart disease includes any heart defects that are present at birth, which are also termed *congenital heart defects*. These defects occur when the heart itself, or the blood vessels near the heart, do not develop normally *in utero*. These include coarctation of the aorta, in which the aorta is abnormally constricted; valvular stenosis, in which one or more heart valves are narrowed; and septal defects, in which the septum or wall separating the right and left sides of the heart is defective, allowing blood from the systemic side to mix with that in the pulmonary side and vice versa.

The remainder of this chapter focuses on the two major diseases of the heart and blood vessels: CAD and HTN.

In Review

- Atherosclerosis is a process in which arteries become progressively narrowed. Coronary artery disease is atherosclerosis of the coronary arteries.
- When coronary artery blood flow is sufficiently blocked, the area of the myocardium supplied by the diseased artery suffers from lack of blood (ischemia). In severe cases, the resulting oxygen deprivation can cause a myocardial infarction, which results in tissue necrosis.

- Atherosclerotic changes in the arteries actually begin in young children, but the extent and progression of this disease process are quite variable.
 - *Hypertension* is the clinical term for high blood pressure.
 - Stroke affects the cerebral arteries so that the part of the brain they supply receives too little blood. Ischemic stroke is the most common form of stroke, usually resulting from cerebral thrombosis or embolism. The other cause of stroke is cerebral hemorrhage (cerebral and subarachnoid).
 - Heart failure is a condition in which the cardiac muscle becomes too weak to maintain an adequate cardiac output, causing blood to back up in the veins.
 - Peripheral vascular diseases involve systemic, rather than coronary, vessels and include peripheral artery disease, varicose veins, and phlebitis.
 - Congenital heart disease includes all heart defects present at birth.
-

Understanding the Disease Process

Pathophysiology refers to the pathology and physiology of a specific disease process or disordered function. Understanding the pathophysiology of a disease gives us insight into how physical activity might affect or alter the disease process. Note that some common underlying pathophysiology such as systemic, subacute inflammation and insulin resistance is associated with many chronic diseases. In the following sections, we examine the pathophysiology of CAD and HTN.

Pathophysiology of Coronary Artery Disease

How does atherosclerosis develop in the coronary arteries? The walls of the coronary arteries are composed of three distinct layers, as shown in [figure 21.3](#): the tunica intima (inner layer), the tunica media (middle layer), and the tunica adventitia (outer layer). These are referred to more simply as the intima, the media, and the adventitia. The innermost layer of the intima is called the **endothelium**. It is formed by a thin lining of endothelial cells that provides a smooth protective coating between the blood flowing through the artery and the intimal layer of the vessel wall. The endothelium provides a protective barrier between toxic substances in the blood and the vascular smooth muscle cells. For vessels larger than 1 mm in diameter, the intima also includes a

subendothelial layer, formed from connective tissue. The media consists mainly of the smooth muscle cells, which control the constriction and dilation of the vessel, and elastin. The adventitia is composed of collagen fibers that protect the vessel and anchor it to its surrounding structure.

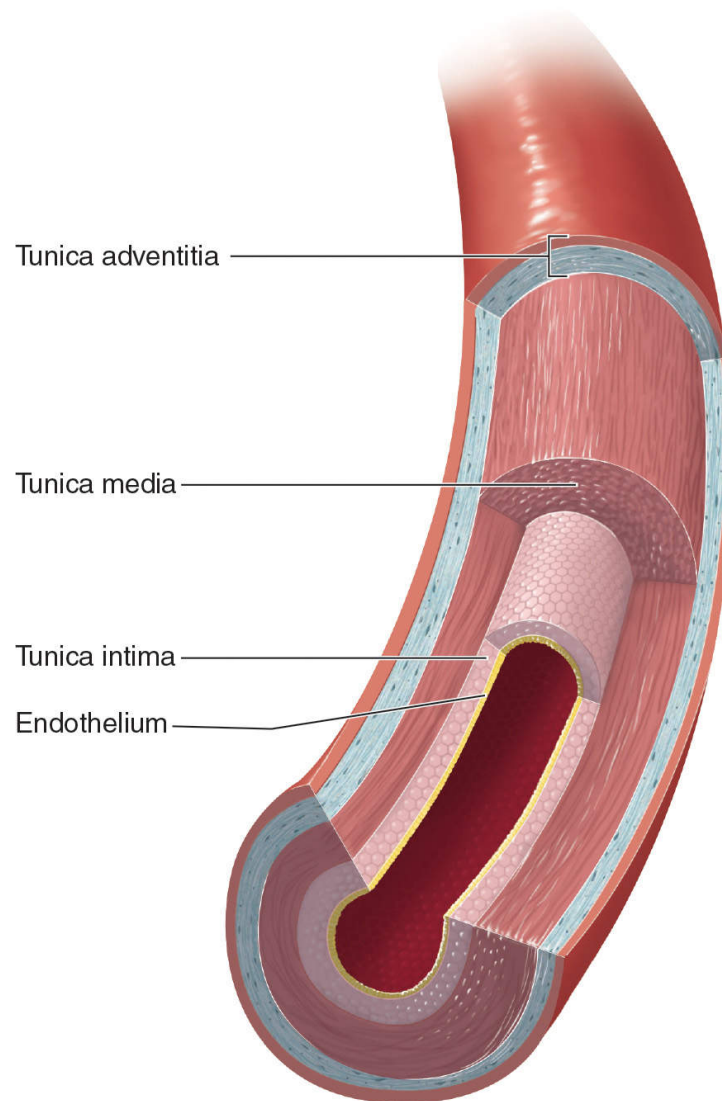


FIGURE 21.3 The wall of an artery has three layers: tunica intima, tunica media, and tunica adventitia.

The endothelium forms a protective barrier between the blood and outward layers of the artery wall. It functions to provide a smooth surface along which blood can travel, but it also has many significant physiological functions, including controlling blood flow within the vascular system by influencing vasodilation and vasoconstriction. It

does this by producing physiologically active substances like **nitric oxide (NO)** and stimulating the production of similar substances in other tissues. These, in turn, cause the smooth muscle within the arterial wall to contract or dilate, thereby exerting control over the flow of blood through those channels.

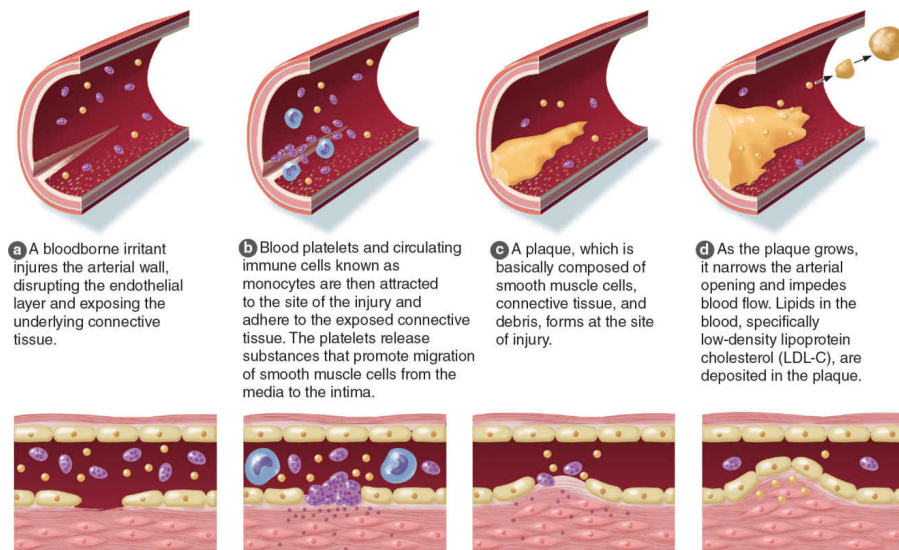


FIGURE 21.4 Changes in the arterial wall (a-d) with injury, illustrating the disruption of the endothelium and the subsequent alterations that lead to atherosclerosis.



Atherosclerosis is an inflammatory disease.²⁹ The initial injury—that is, the event that initiates the onset and progression of atherosclerosis (figure 21.4)—may result from any of several environmental stimuli. Such factors include elevated triglycerides or low-density lipoprotein (LDL) cholesterol after a high-fat meal (postprandial lipemia), compounds in cigarette smoke and inhaled environmental pollutants, **adipokines** and other inflammatory substances associated with excess adipose tissue, infectious microorganisms, and even emotional stress, all of which are associated with vascular inflammation. In turn, the inflammation causes endothelial dysfunction. Platelets in the blood, which may be

attracted to the injury site at the endothelium, adhere to the exposed connective tissue (see [figure 21.4b](#)). These platelets release substances that promote migration of smooth muscle cells from the media into the intima. The intima normally contains few if any smooth muscle cells. A plaque, which is basically composed of smooth muscle cells, connective tissue, and debris, forms at the site of injury (see [figure 21.4c](#)). Eventually, lipids in the blood, specifically LDL cholesterol, are attracted to and deposited in the plaque (see [figure 21.4d](#)). Through this process, the initial injury and the subsequent inflammation result in a local deposition of lipids (fat), smooth muscle cells, and other substances within the wall of the artery.

The inflammatory process is driven by many physiologically active substances produced by the endothelium or actively stimulated by it (or both). These substances include **platelet-derived growth factor (PDGF; figure 21.4b)** and other substances that inhibit or drive endothelial NO production. The exact mechanisms and integration of these factors, as well as their individual contributions to the atherosclerotic process, are complex and still not fully understood.

Low-level, systemic inflammation is an underlying characteristic of many chronic diseases including atherosclerosis, obesity, and HTN (all discussed later in this and the next chapter). Both inflammation and insulin resistance are important forerunners of the pathophysiology of CAD.

Relatively recently, researchers have theorized that plaque forms when monocytes—white blood cells that are effector cells of the immune system—attach between endothelial cells. These monocytes differentiate into macrophages, which ingest oxidized LDL cholesterol. They slowly become large foam cells and form fatty streaks. Smooth muscle cells then accumulate under these foam cells. The endothelial cells then separate or are sloughed off, exposing the underlying connective tissue and allowing platelets to attach to it. The role of monocytes in atherosclerosis is the subject of ongoing research.

The plaque consists of a collection of smooth muscle cells and inflammatory cells (macrophages and T lymphocytes) with both intracellular and extracellular lipid. The plaque also contains a

fibrous cap. The composition of the plaque and of its fibrous cap is critical to its stability. Unstable plaques are those that have thin fibrous caps and are heavily infiltrated by foam cells. These plaques are much more susceptible than others to rupture. When plaques rupture, proteolytic enzymes are released, causing a breakdown of the cellular matrix and leading to blood clotting (thrombus) as illustrated in [figure 21.5](#). The thrombus, depending on its size, can occlude or block the artery, resulting in myocardial infarction. Even if a ruptured plaque does not directly cause an infarction, the thrombus may actually become part of the plaque and further enlarge it. In fact, this may be one reason for the unpredictably progressive nature of the disease. Plaque rupture and thrombosis account for up to 70% of myocardial infarctions and cardiac arrests. Interestingly, the plaques that do rupture are typically small, causing less than 50% stenosis or narrowing of a coronary artery.

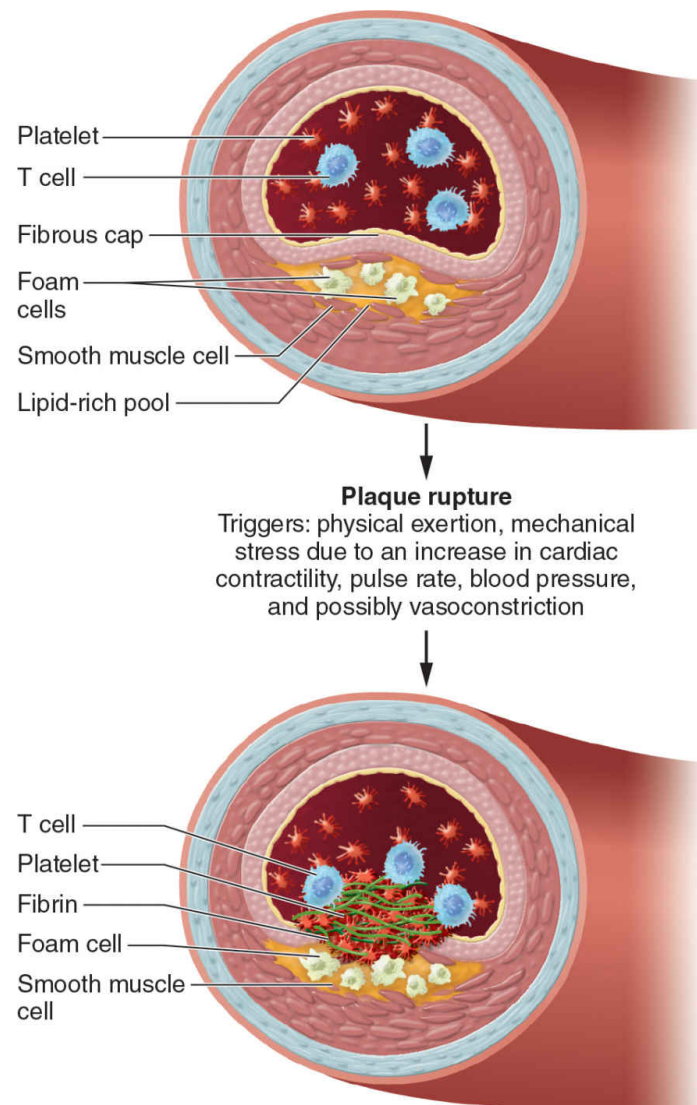


FIGURE 21.5 Illustration of fissure or rupture of an unstable plaque in a coronary artery, releasing its contents into the bloodstream and stimulating the formation of a thrombus (clot).

There is good evidence that a plaque is a dynamic structure, undergoing cycles of erosion and repair that are responsible for its progressive growth. Ironically, smooth muscle cells are important to the stability of the plaque, and smooth muscle cell proliferation is potentially beneficial to maintaining the integrity of the plaque. Plaque rupture sites are characterized by a low density of smooth muscle cells.

Pathophysiology of Hypertension

The pathophysiology of HTN is not well understood. In fact, it is estimated that 90% to 95% of those identified with HTN are

classified as having *essential* HTN, or hypertension of unknown origin. Essential HTN is also referred to as primary or idiopathic HTN. The remaining 5% to 10% are classified as having secondary HTN, meaning that the HTN is secondary to other health issues such as kidney disease, adrenal tumors (that may increase the release of epinephrine), or aortic defects. Although the pathophysiology of the most common type of HTN is largely unknown and undefined, it is clear that it is a multifactorial disease related to hormonal mediators, vascular reactivity (especially elasticity), blood viscosity, and neural stimulation of vascular resistance.

In Review

- *Pathophysiology* refers to the pathology and physiology of a specific disease process or disordered function.
- Early theories held that CHD can be initiated by injury to the smooth endothelial lining of the intimal layer of the arterial wall. This damage causes endothelial dysfunction and also attracts platelets to the area, which in turn release PDGF. The dysfunctional endothelium along with PDGF attracts smooth muscle cells, and a plaque (composed of smooth muscle cells, connective tissue, and debris) begins to form. Eventually, lipids, smooth muscle, collagenous tissue, and even blood clots may be deposited in the plaque.
- More recent research indicates that monocytes, involved with the immune system, can attach between endothelial cells in the intima and begin forming fatty streaks; this also leads to plaque formation. According to this theory, endothelial damage is not necessary for plaque formation, but endothelial dysfunction stimulated by environmental factors may also promote atherosclerosis. The details and role of this process still need to be clarified by further research.
- It is now clear that the composition of the plaque and its fibrous cap is critical with respect to myocardial infarctions and cardiac arrest. Softer plaques, where there is typically less than 50% occlusion of the artery, that have thin fibrous caps and are heavily infiltrated with foam cells are the most unstable and likely to rupture.
- The pathophysiology of HTN is poorly understood.
- More than 90% of people with HTN have essential hypertension, meaning that its cause is unknown.

Cardiovascular Disease Risk

Over the years, scientists have attempted to determine the basic etiology, or cause, of both CAD and HTN. Much of our understanding of these two diseases comes from the field of epidemiology, a science that studies the relationships of various factors to a specific disease or disease process. In a number of studies, selected members of various communities have been observed for extended periods of time. These observations include periodic medical examinations and clinical tests.

Eventually, some of the participants in such studies become diseased and, over time, many die as a result. All those who develop heart disease or HTN or who die from heart attacks or strokes are grouped accordingly. Then their previous medical and clinical tests are analyzed to determine shared attributes or factors. This approach provides researchers with valuable insights into the disease process and may, under some circumstances, provide indirect evidence of the causes of disease.

As identified in long-term longitudinal population studies, the factors that place individuals at risk for disease are referred to as **risk factors**. Often, having more risk factors imparts additional risk because the risk factors are additive and interact; thus, someone with three or four risk factors is at significantly greater risk than someone with one or two risk factors. Let's examine the risk factors for heart disease and HTN.

Risk Factors for Coronary Artery Disease

The factors associated with an increased risk for premature development of CAD can be classified into two groups: those over which a person has no control and those that can be altered through basic changes in lifestyle. Those that a person cannot control include family history of CAD (heredity), race, male sex, and advanced age. According to the American Heart Association (AHA), the **primary risk factors** that can be controlled or altered include

- exposure to or inhalation of tobacco smoke,
- hypertension,
- abnormal blood lipids and lipoproteins,

- physical inactivity,
- obesity and overweight, and
- diabetes.

Table 21.3 shows the levels of some of these risk factors that are associated with increased risk. However, these should be evaluated on an individual basis. One should note that recently published guidelines and practice standards for the management of lipids, HTN, and lifestyle to reduce cardiovascular risk, HTN, and overweight or obesity have significantly changed the practice of risk assessment and altered or even removed the thresholds and optimal values for many of these risk factors.

Other CAD risk factors have been proposed, but there is not yet sufficient evidence to support their inclusion as primary risk factors, that is, factors that add to the ability to predict CAD events. Those listed next have been added to the most recent AHA risk factor assessment guidelines. The AHA recommends that if, based on primary risk factors, a person's risk level is borderline or uncertain, other factors be considered to decide what treatment or changes are appropriate.²⁰

- Family history, specifically first-degree relatives (a parent or sibling), of early CAD (men <55 years or women <65 years).
- C-reactive protein (hs-CRP) levels ≥ 2 mg/L. C-reactive protein is produced in the liver and smooth muscle cells within coronary arteries in response to injury or infection. C-reactive protein is a marker of inflammation.
- High coronary artery calcium (CAC) scoring. Coronary artery calcium is an indicator of coronary artery disease and can be assessed using conventional, noninvasive computed tomography scans. People with CAC ≥ 300 or ≥ 75 th percentile for age, sex, and ethnicity are considered to be at risk.
- Low ankle–brachial index (ABI). This index compares blood pressure taken at the ankle and at the arm (brachial artery). It is a diagnostic test for peripheral artery disease. An ABI <0.9 is considered a risk factor.

Identification of Risk Factors

In July 1948, the National Institutes of Health (National Heart, Lung, and Blood Institute) started the Framingham Heart Study (FHS). The FHS was designed as a longitudinal investigation aimed at identifying those factors that influence the development of cardiovascular diseases. The original study population of 5,209 persons from Framingham, Massachusetts, was examined over a 4-year period starting in September 1948 and was reexamined every 2 years for over 48 years. The FHS pioneered the concept of risk factors associated with the development of CHD. Inclusion of the offspring, or second generation, of the original Framingham group began in 1971, and the third-generation study started in 2002. The FHS has been one of the most successful longitudinal studies in the history of medical research and resulted in over 2,473 research-based publications as of 2012. As just one example, the FHS was the first to indicate the importance of cholesterol as a risk factor for CHD, and subsequently to demonstrate that the true risk was associated with high levels of LDL cholesterol and low levels of high-density lipoprotein (HDL) cholesterol.

TABLE 21.3 Level of Risk for Coronary Artery Disease Associated with Selected Risk Factors

Risk factor		Level of risk		
		Optimal	Borderline	At risk
Blood pressure^a				
Systolic	mmHg	<120		>140
Diastolic	mmHg	<80		>90
Overweight-obesity (body mass index) ^a	kg/m ²	18.5-24.9	25.0-29.9	> 30
Fasting plasma glucose ^b	mg/dl	<100	100-125	>126
Physical activity ^{a,c}	min/week	150-300		<150

^aAmerican Heart Association, 2010.

^bData from the American Diabetes Association: www.diabetes.org/diabetes-basics.

^cModerate to vigorous exercise on most days of the week.

The Bogalusa Heart Study is another long-term epidemiological study (similar to the FHS) that began in 1972 and continues today. The Bogalusa study is a longitudinal study of cardiovascular disease risk factor development from birth through age 39. In 204 of the subjects who died prematurely (primarily from accidents, homicides, or suicides), the scientists found a strong relation between the risk

factors and development of fatty streaks; the greater the number of risk factors, the greater the development of aortic and coronary artery fatty streaks.⁵ In fact, the authors of the Bogalusa Heart Study were among the first to describe the presence of fatty streaks, the origins of atherosclerotic heart disease, in children.

Lipids and Lipoproteins

For many years, cholesterol and triglycerides were the only lipids observed in epidemiological studies. More recently, scientists have studied the manner in which lipids are transported in the blood. Lipids are insoluble in blood, so they are packaged with a protein to allow transport through the body. **Lipoproteins** are the proteins that carry the **blood lipids**. Two classes of lipoproteins of major concern for CHD are **low-density lipoprotein cholesterol (LDL-C)** and **high-density lipoprotein cholesterol (HDL-C)**. Low-density lipoprotein cholesterol has been implicated in plaque formation, whereas HDL-C is probably involved in plaque regression. High levels of LDL-C and low levels of HDL-C place a person at increased risk of having a heart attack. Conversely, a high level of HDL-C and a low level of LDL-C indicate a lower risk for heart attack. Thus, the ratio of the two lipoproteins is important. Simply assessing total cholesterol is not adequate to predict risk. A person with a moderately high level of total cholesterol (Total-C) coupled with a high concentration of HDL-C and low concentration of LDL-C is at a lower risk than a person with a moderately low level of Total-C but a higher level of LDL-C and a low HDL-C. Because of the important and distinct functions of these two lipoproteins, physicians and researchers now use the term *dyslipidemia* to refer to the lipoprotein-related risk factor for CHD, rather than “high cholesterol.” The emphasis is on the relative levels of LDL-C and HDL-C.

Why are these two cholesterol carriers associated with different risk levels? Low-density lipoprotein cholesterol is thought to be responsible for depositing cholesterol in the arterial wall, whereas HDL-C removes cholesterol from the arterial wall and transports it to the liver to be metabolized. Because of these very different roles, it is essential to know the specific concentrations of both of these lipoproteins when one is determining individual risk. The new

AHA/ACC (American College of Cardiology) guidelines for managing dyslipidemia no longer specify target treatment levels for lipids, but rather assess the need to treat with drugs and the effectiveness of any treatment based on the levels of LDL-C.³⁸ The preferred treatment strategy involves prescribing HMG-CoA reductase inhibitors, or statins. These are a class of drugs that block the formation of substances used to produce cholesterol in the liver. Thus, unlike the previous treatment guidelines (ATP3), the new guidelines give no specific targets for LDL, HDL, or even ratios of lipoprotein classes; rather, the therapy is guided by its effectiveness.

RESEARCH PERSPECTIVE 21.2

Maintaining Fitness Into Middle Age Reduces CVD Risk

Aerobic fitness has a dose–response relation with cardiorespiratory fitness, which confers many health benefits, including overall lower risk of cardiovascular disease, metabolic syndrome, diabetes, and total mortality. Studies that have examined the effects of changes in cardiorespiratory fitness suggest that those who maintain or improve fitness over time have lower risk for developing and dying of cardiovascular diseases, including hypertension, high cholesterol, and diabetes, compared to peers with longitudinal decreases in fitness. However, most of these longitudinal studies have focused on a white male population, reducing their generalizability, and have typically spanned only a 5- to 7-year period.

In 2015, an analysis of participants in the Coronary Artery Risk Development in Young Adults (CARDIA) study examined how changes in fitness over 20 years affected cardiometabolic risk factors in middle age.¹¹ The CARDIA study is an ongoing study of African American and Caucasian men and women that has collected fitness and cardiovascular health data over 25 years. For this investigation, the study team was interested in how fitness changed over the first 20 years of the study and how those changes in fitness were related to cardiometabolic outcomes. They used multiple linear regression and proportional hazards regression with adjustments for individual differences in baseline values to analyze data from 2,048 participants (43% men, ~50% African American). Across the 20-year span of the study, 20.6% of participants maintained their fitness level. The regression analysis revealed that maintaining fitness was associated with greater increases in high-density lipoproteins (HDLs) and smaller increases in low-density lipoproteins (LDLs) and triglycerides. Participants who maintained fitness had also gained less weight and waist circumference after 20 years. Overall, maintaining fitness over 20 years is associated with more favorable cardiometabolic risk factors in middle age. These findings extend previous

findings to a larger and more generalizable cohort and underline the importance of maintaining fitness throughout the life span.

A new, long-awaited set of practice guidelines for the treatment of high cholesterol was published in 2013,³⁸ replacing previous guidelines from 2001. The ATP4 (Adult Treatment Panel) guidelines take an entirely different (and somewhat controversial) approach to treating high cholesterol. There are no overtly expressed treatment goals based on LDL or total cholesterol concentrations; rather, the recommendation is risk reduction through identification of statin benefit groups. These are groups of individuals who would benefit from pharmacological treatment with statins. People who fall within one of these specified groups should be treated with moderate- or high-intensity statin therapy. The benefit groups are identified according to risk as individuals

- with clinically diagnosed CHD;
- without clinically diagnosed CHD but with LDL ≥ 190 mg/dl;
- not in the preceding category, but between 40 and 75 years of age with type 1 or type 2 diabetes; and
- not in the preceding categories but aged 40 to 75 years with a 10-year risk $\geq 7.5\%$ (by new risk assessment standards that use a complex formula based on age, sex, total and HDL cholesterol, smoking status, SBP, and use of antihypertensive medications).

The controversy about this particular set of guidelines arises from the potential increase in the use of statins among Americans, even those who do not have elevated cholesterol concentrations. Some estimate that statin prescriptions will double, increasing the use from 15 million to 30 million Americans. The cost of individual statin drugs ranges from about \$0.50 to \$4.00 per dose, depending on which statin is prescribed.

Important for exercise professionals, the expert panel prefaces the new guidelines with the following statement:

It must be emphasized that lifestyle modification (i.e., adhering to a heart healthy diet, regular exercise habits, avoidance of tobacco habits, and maintenance of a healthy weight) remains a critical component of health

promotions and ASCVD risk reduction, both prior to and in concert with the use of cholesterol-lowering drug therapies.

Risk Factors for Hypertension

The risk factors for HTN, like those for CHD, can be classified as those we can control and those we cannot. Those we cannot control are heredity (family history of HTN), sex, advanced age, and race (increased risk for people of African or Hispanic ancestry). Risk factors we can control are

- obesity and overweight,
- diet (sodium, alcohol),
- use of tobacco products,
- use of oral contraceptives,
- stress, and
- physical inactivity.

Although heredity is a risk factor for HTN, it probably plays a much smaller role than many of the other proposed factors. We must remember that lifestyle factors are often quite similar within a family.

Recently, scientists have shown great interest in a possible link between HTN, obesity, type 2 diabetes, and CHD through the common pathways of inflammation and insulin resistance. Obesity also has been established as an independent risk factor for HTN. Numerous studies have shown substantial reductions in blood pressure with weight loss in hypertensive patients. Also, although sodium intake traditionally has been linked to HTN, this relationship is controversial.

Physical inactivity is a risk factor for HTN. Its role has been conclusively established in epidemiological studies. Furthermore, substantial evidence indicates that increasing physical activity tends to reduce elevated blood pressure.

In Review

- Risk factors for CHD that we cannot control are heredity (and family history), male sex, and advanced age. Those that we can control are abnormal blood lipids and lipoproteins, HTN, smoking, physical inactivity, obesity, and diabetes.

- Inflammation, insulin resistance, and vascular dysfunction are common pathophysiological conditions found in atherosclerosis, type 2 diabetes mellitus, obesity, HTN, and other chronic diseases.
 - Low-density lipoprotein cholesterol is thought to be responsible for depositing cholesterol in the arterial walls. However, HDL-C acts as a scavenger, removing cholesterol from the vessel walls. Thus, high HDL-C levels provide some degree of protection from CHD.
 - Risk factors for HTN that cannot be controlled include heredity, advanced age, and race. Those we can control are obesity, diet (sodium and alcohol), use of tobacco products and oral contraceptives, stress, and physical inactivity.
-

Reducing Risk Through Physical Activity

The potential role of physical activity in preventing or delaying the onset of CHD and HTN has been of major interest to the medical community for many years. In the following sections, we try to unravel this mystery by examining the following areas:

- Epidemiological evidence
 - Physiological adaptations with training that might reduce risk
 - Risk factor reduction with exercise training
-

RESEARCH PERSPECTIVE 21.3

Physical Activity Reduces Cigarette Cravings

Cigarette smoking is a global health concern. In the United States, over 18% of adults smoke cigarettes and about one in five deaths is caused by smoking. Many smokers want to quit but only 3% to 5% of unaided attempts are successful after 1 year. Most people who quit smoking relapse within the first 8 days. Even with assistive behavioral or pharmacological therapy, fewer than 30% of smokers successfully quit smoking. Physical activity has been recommended as an aid for smoking cessation programs; however, there is little scientific evidence of the ability of physical activity to assist in smoking cessation.

Recently, an individual patient data meta-analysis investigated the acute effects of short bursts of physical activity on cigarette craving.¹⁸ The meta-analysis reviewed 19 intervention studies that had examined the acute effects of physical activity on the desire to smoke. The interventions included moderate-intensity walking, running, light to vigorous cycling, and isometric exercise, with exercise bouts lasting anywhere from 5 to 40 min. Short bouts

of physical activity resulted in large reductions in the desire to smoke. Furthermore, the effect size of the physical activity intervention was much larger when moderate-intensity exercise was considered, suggesting that the intensity of the exercise is likely related to the acute reduction in cravings. The researchers concluded that acute bouts of physical activity reduce cigarette cravings in individuals who are trying to quit smoking. Future studies should investigate if the mode, intensity, or duration of the activity can modulate the reduction in cravings and should determine if acute reductions in cravings can lead to long-term success in quitting smoking.

Reducing the Risk of Coronary Artery Disease

Physical activity has been proven effective in reducing the risk of cardiovascular disease. In the following sections, we discover what is known about this topic and what physiological mechanisms are involved.

Lifestyle Management to Reduce the Risk of Cardiovascular Disease

New practice guidelines have been established by the AHA and the ACC to assist physicians and other health care professionals in evaluating the roles of physical activity and a healthy diet in the prevention and treatment of cardiovascular disease (CVD).¹⁴ These are evidence-based guidelines focused on answering critical questions, an approach similar to that of practice guidelines and standards published jointly by the AHA and the ACC in 2013.^{20,38} In the evidence-based approach, recommendations are based largely on analyses of randomized controlled trials (RCT) and on meta-analyses of those trials. The evidence present in the research is classified as strong, moderate, weak, or insufficient. (A final category, “recommend against,” is also sometimes used.) The resulting guidelines replace similar previously published guidelines and practice standards.

The three critical questions addressed by the CVD guidelines are as follows:

- What are the effects of dietary pattern and macronutrient composition on risk factors and outcomes for CVD when compared to no treatment and to other types of treatment?

- What are the effects of dietary intake of sodium and potassium on CVD risk factors and outcomes when compared to no treatment and to other types of treatment?
- What is the effect of physical activity on blood pressure and serum lipids compared to no treatment and to other types of treatment?

Based on the existing research literature, the following summarized conclusions were reached:

Dietary pattern. This question centers on the efficacy of specific dietary changes for controlling lipids (specifically LDL cholesterol) and blood pressure. Strong evidence supported a dietary pattern emphasizing eating a largely plant-based diet (vegetables, fruits, and whole grains), low-fat dairy products, poultry, and fish and restricting sugar-sweetened beverages and red meats, as well as restricting saturated fat intake to 5% to 6% of total calories and reducing intake of trans fatty acids.

Dietary sodium and potassium. This question focuses on adults who may benefit from lowering resting blood pressure. Specifically, it highlights two micronutrients: sodium and potassium. The recommendations include a dietary pattern similar to that recommended for the first critical question. Strong evidence supported sodium restriction, either limiting sodium to the recommended daily intake (RDI) of 2,400 mg/day or reducing sodium intake by 1,000 mg/day if the RDI cannot be reached. A further reduction in sodium intake to 1,500 mg/day, if possible, is associated with further reductions in blood pressure and is recommended.

Physical activity. This question centers on the effects of exercise and physical activity on lowering lipids and blood pressure. Moderate evidence supported a role for aerobic exercise in lowering LDLs and total cholesterol but not in altering triglycerides or HDLs. The statement recommends aerobic activity for three or four sessions per week, 40 min per session, at a moderate to vigorous intensity. The evidence was strong in supporting aerobic exercise for the reduction of blood pressure for those with HTN, and similar exercise recommendations are suggested.

The guidelines recommend similar dietary changes and exercise for all adults with the goal of reducing risk factors, including 150 min/week of moderate-intensity exercise or 75 min/week of vigorous-intensity exercise. There is little doubt that a healthy dietary pattern and daily physical activity are healthy partners in a lifestyle to prevent chronic disease.

Inactivity as a Risk Factor: Epidemiological Evidence

Hundreds of research papers have dealt with the epidemiological relationship between physical inactivity and CAD. Generally, studies have shown the risk of heart attack in sedentary male populations to be about two to three times that of men who are physically active in either their jobs or their recreational pursuits. Early epidemiological studies from the 1950s were among the first to demonstrate this relation.³² In these studies, sedentary bus drivers were compared with active bus conductors who worked on double-decker buses, and sedentary postal workers were compared with active postal carriers who walked their routes. The death rate from CAD was about twice as high in the sedentary groups as in the active groups. Many studies published over the subsequent 20 years showed essentially the same results: People who were occupationally sedentary were at approximately twice the risk for death from CAD as those who were active.

Most of these early epidemiological studies focused exclusively on occupational activity. Not until the 1970s did researchers start looking at leisure-time activity as well. Dr. Morris and his colleagues, who conducted the early studies of bus drivers and postal workers, were among the first to observe the relationship between leisure-time activity and the risk of CAD: The least active people were at two to three times greater risk. Subsequent studies have provided similar results.⁸ Physical inactivity approximately doubles the risk of having a fatal heart attack. While most of these early studies were conducted on men, subsequent investigations have demonstrated similar results in women.¹²

Many extensive reviews of epidemiological studies published on physical inactivity and CAD have been published in the past 30 to 40 years. Many have applied stringent criteria for inclusion of studies

and the quality of the research. Most indicate that inactive people have about twice the risk of dying from heart disease as more active people. Additionally, these studies have demonstrated that the relative risk from physical inactivity is similar to the risk associated with other major risk factors for CAD.² The results from these epidemiological studies have played a major role in leading the AHA and other organizations to declare physical inactivity a primary risk factor for CAD.

Exercise Type and Intensity

Another important concern was raised in the mid-1980s: What level of physical activity or fitness is necessary to reduce one's risk for CAD? It was not totally clear from the epidemiological studies what threshold level of fitness or activity was effective in reducing risk. In fact, during the mid-1980s, scientists had just started to differentiate between activity level (a pattern of behavior) and fitness (defined by a person's $\dot{V}O_{2\max}$), a discussion that continues today.^{7,33} In retrospect, distinguishing these two terms was crucial because a person can be active yet unfit (low $\dot{V}O_{2\max}$) or fit (high $\dot{V}O_{2\max}$) yet inactive. This area of research was subsequently redirected based on various epidemiological studies demonstrating that the levels of activity associated with a lower risk for CAD were generally low, well below the level that would be necessary to increase aerobic capacity. Subsequent studies have supported this concept.⁸ Low levels of activity, such as walking and gardening, can provide considerable benefit in reducing the risk for CAD. However, it has also been demonstrated that more vigorous exercise may provide even greater benefit.

In 2002, a group of scientists from Harvard University reported the relationship of exercise type and intensity to CAD in more than 44,000 men enrolled in the Health Professionals Follow-Up Study.³⁹ These men were followed every 2 years from 1986 through 1998 to assess potential CAD risk factors, identify newly diagnosed cases of CAD, and assess levels of leisure-time physical activity. Men who ran 6 mph (9.7 km/h) or faster for 1 h or more per week had a 42% risk reduction compared with men who didn't run. Men who trained with weights for 30 min or more per week had a 23% risk reduction

when compared with men who didn't weight train. Brisk walking for 30 min or more per day was associated with an 18% risk reduction, as was rowing for 1 h or more per week. Surprisingly, swimming and cycling were unrelated to risk. This study was the first to show the direct benefits of weight training on CAD risk and indicate that exercise intensity is also a critical consideration, with higher intensities providing greater risk reduction.

More recently, research has shown that the type and intensity of exercise is directly related to mortality and disease prevention.⁴² The main outcome of this research was that people who didn't meet the physical activity guidelines recommended by the American College of Sports Medicine (ACSM) (150-250 MET-minutes per week) had a significantly higher mortality than those who met or exceeded those guidelines. In fact, after adjustment for medication use and body mass index, and even such variables as education and dietary pattern, those who exceeded the recommendations by two to three times had almost 25% lower mortality rates than those who just met the recommendations. There was also a modest dose-response relation between the amount of walking and the reduction in mortality⁴² (figure 21.6).

Effects of Exercise on Risk Factors

The importance of regular physical activity and exercise in reducing the risk of CAD becomes apparent when we consider anatomical and physiological adaptations in response to exercise training. Many studies have investigated the role of exercise in altering risk factors associated with heart disease. Let's consider the major risk factors and how exercise might affect them.

Acute exercise and even lower levels of physical activity that may not induce significant training effects can directly and indirectly affect the risk for and the progression of CAD. Such basic and important factors as inflammation, endothelial function, insulin resistance, the production and secretion of adipokines, glucose metabolism, blood pressure, and even the influence of lipoproteins in the blood are positively influenced by exercise and physical activity.

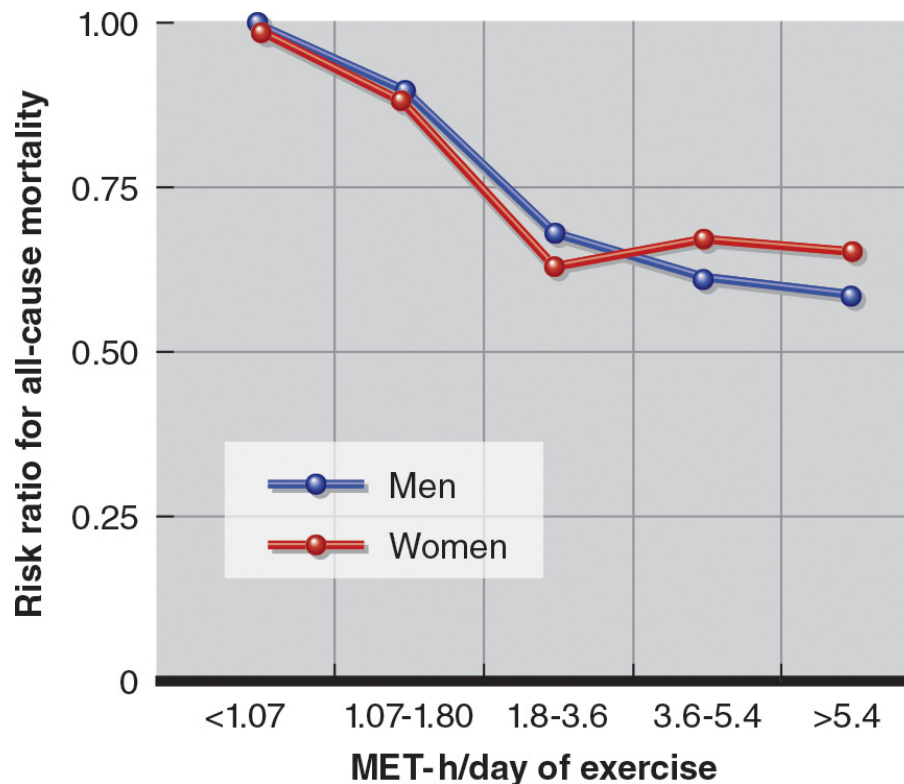


FIGURE 21.6 The relation between total daily exercise and all-cause mortality for men and women in the Williams study. This is a dose–response curve, indicating that the higher the dose, the better the response.

Data from Williams (2013).

Smoking There is little direct evidence to indicate that exercise leads to smoking cessation or reduces the number of cigarettes smoked.

Lipids According to the recently published AHA guidelines on lifestyle management, there is moderately strong evidence that aerobic (endurance) exercise decreases LDL-C, but the evidence for the effects of exercise in altering either triglycerides or HDL-C is low.¹⁴ While many studies have demonstrated increases in HDL-C and decreases in triglycerides from training, others have reported little or no change.

Postprandial (after a meal) lipemia, the influx of fat into the blood after a fatty meal, increases the risk for CAD, stimulates plaque progression, and decreases plaque stability. Exercise before a high-fat meal moderates this effect by decreasing the amount of lipids entering the blood. High fat (or even moderate amounts of fat) in a

meal can cause a significant postprandial outflow of lipoproteins that is atherogenic and, in fact, a risk factor.

We must consider two confounding factors when we evaluate lipid changes with exercise training because each can have a marked effect. Because plasma lipids are expressed as a concentration (milligrams of lipid per deciliter of blood), any change in plasma volume will affect plasma concentrations independently of the change in total lipid. Recall that training typically increases plasma volume (discussed in [chapter 11](#)). With this plasma expansion, the absolute amount of HDL-C could increase yet the HDL-C concentration might not change or even could decrease. In addition, plasma lipid levels are tightly coupled with changes in body weight. When we evaluate the effects of exercise training, the independent effects that a change in body weight could have on plasma lipids must be considered.

Hypertension Relatively strong data support the effectiveness of exercise in reducing blood pressure in those with mild to moderate HTN. Endurance training can reduce systolic (2-5 mmHg) and diastolic blood pressures (1-4 mmHg) in all individuals, including those with HTN.¹⁴ The specific mechanisms responsible for the decreases in blood pressure with endurance training have yet to be fully determined.

Insulin Sensitivity and Glucose Exercise improves glucose uptake and utilization, as well as insulin sensitivity.¹⁷ In fact, a single bout of exercise has been shown to improve insulin receptor sensitivity.

Endothelial Function Strong evidence demonstrates that exercise training improves endothelial function. Endothelial dysfunction is primarily the result of decreased NO bioavailability. Exercise training has been shown to increase NO bioavailability.⁴⁰

Inflammation Moderate exercise is acutely anti-inflammatory.¹⁹ Exercise has been shown to be associated with decreased levels of inflammatory markers and with increased immune system substances that slow the atherosclerotic process.

Other Factors With respect to the remaining risk factors, exercise plays an important role in weight reduction and control and in the

management of diabetes. These topics are discussed in detail in [chapter 22](#). Exercise also has been reported to be effective for reducing and controlling stress, reducing anxiety, and treating depression and anxiety.⁹

Endurance training produces other, long-term favorable anatomical and physiological changes that decrease the risk of heart attack, including larger coronary arteries, increased heart size, and increased pumping capacity. It also has a favorable effect on most of the risk factors for CAD. While not studied to the same extent, resistance training appears to provide many of these same benefits. This may be especially true when one combines aerobic training with resistance training, which may positively affect many other physiological variables pertinent to primary and secondary disease prevention.⁴¹

Sedentary Time and Disease Risk

In [chapter 20](#), we mentioned sedentary death syndrome, a term coined by Dr. Frank Booth and colleagues. They have studied the physiology (and genetics) of a sedentary lifestyle, postulating that humans are genetically driven to be physically active and that being sedentary is a root cause of all chronic disease and pathophysiological dysfunction. More recently, an emerging research trend has been to look at sedentary time as reflected by the amount of time populations spend watching TV, working, playing or watching videos on a computer, and engaging in other seated activities. Most of us spend large amounts of time sitting—some more sedentary people may spend up to 12 to 15 h per day sitting.

The epidemiological research related to sedentary time and mortality from heart disease is sparse and not well developed. It is clear that sedentary time confers increased risk for fatal and nonfatal CVD.¹⁶ This appears to be true regardless of whether someone is a regular exerciser or not.¹³ The time spent sitting or engaging in a sedentary activity (irrespective of sleep time) appears to be a significant factor in a healthy lifestyle.

The National Health and Nutrition Examination Survey (NHANES) database (an ongoing analysis of a cross section of U.S. residents) provides further support for the direct relation between the amount of

sedentary time and such risk indicators as HDL-C, triglycerides, insulin and insulin resistance, and inflammatory markers. Longer periods of sitting and higher daily sitting times are associated with harmful metabolic changes across the board, irrespective of whether moderate or vigorous exercise is interspersed.²⁶

Reducing the Risk of Hypertension

Physical activity's role in reducing the risk of developing HTN has not been as well established as its role with respect to CAD and its role in lowering already elevated blood pressure. As we saw in the previous section, exercise training lowers blood pressure in those with moderate HTN, but the precise mechanisms allowing this reduction are not yet fully known. Let's consider what is known.

Epidemiological Evidence

Few epidemiological studies have dealt with the relation between physical inactivity and HTN. In the Tecumseh Community Health Study, 1,700 men (age 16 and older) completed questionnaires and interviews to provide estimates of their average daily energy expenditures, their peak daily energy expenditures, and the hours they spent in particular activities. The more active men had significantly lower systolic and diastolic blood pressures, irrespective of age.³¹ In a follow-up of the participants from the Cooper Clinic study, the investigators reported a relative risk of 1.5 for the development of HTN in people with low levels of fitness compared with highly fit people.⁶ The NHANES database revealed that newly identified HTN was associated with low fitness estimated from a submaximal treadmill test. Odds ratios of 2.12 for women and 1.83 for men (20-49 years) indicated that low fitness levels more than double the risk of HTN in women and almost double the risk in men.¹⁰ (An odds ratio represents the odds that an outcome will occur given a particular risk factor. An odds ratio greater than 1.0 implies an association between the outcome and the risk factor.)

These limited studies indicate that active people and fit people are at reduced risk for developing HTN. Epidemiological studies also have shown that higher physical activity levels and aerobic fitness are related to a decreased risk for stroke in both men²⁸ and women.²³

Training Adaptations That Might Reduce Risk

Specific mechanisms responsible for reductions in resting blood pressure with endurance training have not been clarified. The most likely mechanisms include neurohumoral adaptations, decreased sympathetic nervous system activity, changes in the renin–angiotensin system (a critical control system for blood pressure), and perhaps even structural changes in the vascular system itself.

The phenomenon known as postexercise hypotension (PEH) was noted almost 50 years ago. Subsequently, it has been confirmed in most exercising populations, and the evidence is overwhelming that PEH is a significant, acute effect of exercise that may in fact be linked to the chronic effect of exercise on blood pressure in people with HTN. Postexercise hypotension occurs in people with normal blood pressure as well as in people with HTN. It lasts from 1 to 2 h postexercise and sometimes up to 24 h postexercise. The greatest reductions in blood pressure seem to occur in those with the highest preexercise blood pressure. Reductions in SBP of 4 to 15 mmHg have been reported. The fact that this is an acute response that may last up to 24 h further supports the need for regular, daily physical activity, especially in those with HTN.

Risk Reduction with Exercise Training

In the previous section on CHD, we saw that exercise training lowers resting blood pressure in normotensives and in those with HTN. These reductions are unrelated to the duration of the training program but might be greater in response to low- to moderate-intensity activity compared with higher-intensity activity.

Not only does exercise reduce blood pressure in itself but also it affects other risk factors. Exercise is important in reducing body fat, can increase muscle mass, and promotes improved glycemic (blood sugar) control. Exercise training also has been associated with stress reduction.

In Review

- Epidemiological studies generally have found that the risk of CAD in sedentary male populations is about two to three times that of men who are physically

active, and that physical inactivity approximately doubles a person's risk of a fatal heart attack.

- The levels of activity associated with a reduced risk for CAD can be lower than those needed to increase aerobic capacity.
 - Physical training improves the heart's contractility, work capacity, and coronary circulation.
 - Exercise has a significant impact on LDL-C. Endurance training decreases the ratios of LDL-C to HDL-C and of Total-C to HDL-C.
 - Exercise is anti-inflammatory and improves endothelial function.
 - Exercise can help control blood pressure, weight, and blood glucose levels and alleviate stress.
 - People who are active and those who are fit have reduced risk for developing HTN.
 - Exercise helps to reduce blood pressure in people with HTN.
 - Resting blood pressure is decreased by training in individuals both with and without HTN; this is probably attributable to decreased peripheral resistance, coupled with neurohormonal mechanisms, but the actual mechanisms are not well defined.
 - Exercise also reduces body fat, blood glucose levels, and insulin resistance, factors related to an increased risk for HTN.
-

Risk of Heart Attack and Death During Exercise

When someone dies while exercising, the incident usually makes headlines. Deaths during exercise don't happen often, but they are highly publicized. The stories at the beginning of this chapter offer a few examples. How safe, or how dangerous, is exercise? Death rates and mortality associated with exercise and physical activity among adults (not including youth and adult athletes) have been documented in the scientific literature. The incident rates depend on how the numbers are expressed, and they vary according to population (e.g., healthy adults, those with diagnosed CAD) and sex (women have higher rates than men for unknown reasons).¹ As summarized by the ACSM,¹ these rates vary from 1 cardiovascular event in almost 3,000,000 person-hours of activity, according to the YMCA data, to 1 fatality per 396,000 person-hours of jogging, to 1

fatality in 2.597 million workouts reported by a commercial health-fitness facility. Fatality rates in medically supervised exercise programs for patients with diagnosed CAD have been reported as 1 in 752,365 patient-hours. The extremely low rates in this population are undoubtedly the result of the medical supervision, available emergency equipment and assistance, and the pre-entry medical screening.

The overall risk of heart attack and death during exercise is very low. Further, although the risk of death increases during a period of vigorous exercise, habitual vigorous exercise is associated with an overall decreased risk of heart attack.^{1,37} This is illustrated in [figure 21.7](#). There is concern, however, regarding athletes who pursue ultra-endurance exercise, that is, training bouts or competition in excess of 4 h per session. Theoretically, they are at greater risk for cardiovascular disorders because of the high oxidative stress associated with this type of training or competition. Additionally, a recent study of ultra-endurance athletes (runners) found that “long-term excessive sustained exercise may be associated with coronary artery calcification, diastolic dysfunction, and large-artery wall stiffening” (p. 587).³⁴ This conclusion was somewhat speculative and requires further documentation. It is important to note, as the authors also state, that runners live longer than nonexercisers and have excellent physical fitness levels, which are also associated with lower mortality.

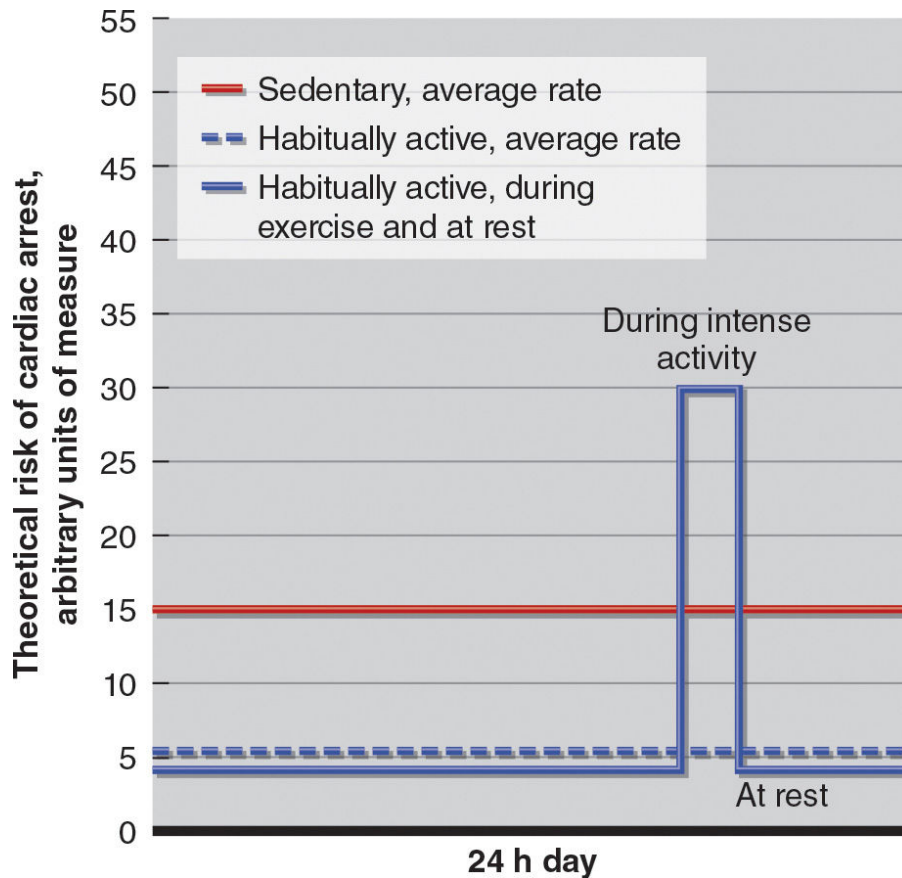


FIGURE 21.7 The risk of primary cardiac arrest during vigorous exercise and at other times throughout a 24 h period, comparing sedentary men with habitually active men.
Data from Siscovick et al. (1984).

When death during exercise occurs in people age 35 or older, it usually results from a cardiac arrhythmia caused by atherosclerosis of the coronary arteries. On the other hand, those under age 35 are most likely to die from hypertrophic cardiomyopathy (enlarged diseased heart, usually genetically transmitted), congenital coronary artery anomalies, aortic aneurysm, or myocarditis (inflammation of the myocardium).

The risk of heart attack increases during the actual period of exercise. However, over the course of a 24 h period, those who exercise regularly have a much lower risk of a heart attack than those who do not exercise.

Exercise Training and Rehabilitation of Patients with Heart Disease

Can active participation in a cardiac rehabilitation program that has a strong aerobic and resistance exercise component help a heart attack survivor to either survive a subsequent attack or avoid one altogether? Endurance training leads to many physiological changes that reduce the work or oxygen demand of the heart and can be seen as preventive for future events. As we have seen, many of these are peripheral and do not involve the heart directly but are directly beneficial to the prevention of the pathophysiology of chronic diseases, specifically CAD. Exercise training increases the capillary/skeletal muscle fiber ratio and plasma volume and improves the efficiency of the working heart, thus resulting in a decreased workload on the left ventricle. Endurance training also improves vascular endothelial function, decreases insulin resistance through increased insulin receptor sensitivity, and increases glucose utilization; exercise training is anti-inflammatory and moderates postprandial lipemia (fat in the blood after a high-fat meal). All of these changes promote cardiovascular health and contribute to the preventive effects of regular exercise training.

VIDEO 21.1 Presents Ben Levine on the clinical implications of $\dot{V}O_{2\max}$.



Significant changes also may occur in the heart itself. Studies of heart disease patients at Washington University in St. Louis have provided dramatic evidence that high-intensity aerobic conditioning not only can substantially change peripheral factors but also can

alter the heart itself, possibly increasing blood flow to the heart and increasing left ventricular function.

From our previous discussions in this chapter, it is clear that endurance exercise training can significantly reduce the risk of cardiovascular disease through its independent effect on the individual risk factors for CAD and HTN. Favorable changes in blood pressure, lipid levels, body composition, glucose control, and stress have been reported in patients undergoing exercise training for cardiac rehabilitation. We have every reason to believe that these changes are just as important to the health of a patient who has had a heart attack as they are for an apparently healthy person.

Resistance training also has substantial benefits when included as a part of a comprehensive cardiac rehabilitation program.⁴¹ [Table 21.4](#) provides a summary comparison of the benefits of resistance training and aerobic training on various physiological and clinical markers of health and fitness. It is obvious that combining the two types of training into a comprehensive rehabilitation program will maximize the overall training benefits.

A comprehensive cardiac rehabilitation program must consider all aspects of the patient's recovery, not just exercise and physical activity. Nutrition counseling is extremely important for everyone, dealing not only with the total calories consumed in a day but most importantly with food selection and dietary pattern to minimize risk and maximize health. Psychological and sexual counseling might also be necessary in some patients. It is not unusual for patients to become anxious about their heart, and their spouses sometimes fear that having sexual relations will further harm the heart of their recovering husband or wife. Many cardiac rehabilitation programs have patient support groups in which patients can openly discuss their concerns.

Patient education, however, is not (and cannot be) the sole answer to helping patients change behavior. Clearly, providing information does not cause lifestyle change nor is it even associated with long-term changes in an individual's health habits. Disease management programs that focus on health behavior change are becoming integral to rehabilitation programs for all kinds of chronic disease, including obesity, diabetes, heart disease, and chronic lung

disease. Such programs require intensified patient contact, tools for behavior change and relapse prevention, and lots of support from within and particularly from outside the health care environment.²¹

TABLE 21.4 Comparison of the Effects of Aerobic Endurance Training and Strength Training on Health and Fitness Variables

Variable	Aerobic exercise	Resistance exercise
Bone mineral density	↑	↑↑
Body composition		
% fat	↓↓	↓
Lean body mass	↔	↑↑
Strength	↔	↑↑↑
Glucose metabolism		
Insulin response to glucose challenge	↓↓	↓↓
Insulin sensitivity	↑↑↑↑	↑↑↑↑
Serum lipids		
HDL-C	↑↔	↑↔
LDL-C	↓↔	↓↔
Resting heart rate	↓↓	↔
Stroke volume	↑↑	↔
Blood pressure at rest		
Systolic	↓↔	↔
Diastolic	↓↔	↓↔
$\dot{V}O_{2\max}$	↑↑↑	↑↑
Endurance performance	↑↑↑	↑↑
Basal metabolism	↑	↑↑

Note. HDL-C = high-density lipoprotein cholesterol; LDL-C = low-density lipoprotein cholesterol; ↑ = increase; ↓ = decrease; ↔ = little or no change. The more arrows, the greater the change.

Reprinted from M.L. Pollock and K.R. Vincent, "Resistance Training for Health," *Research Digest: Presidents' Council on Physical Fitness and Sports* 2, no. 8 (1996): 1-6.

A number of researchers have tried to determine whether participation in a cardiac rehabilitation program reduces the risk of a subsequent heart attack or of death from a subsequent heart attack. It is nearly impossible to design a study to answer these questions, primarily because it would be necessary to enroll several thousand people into one study to have a large enough sample to prove a statistically significant effect. Consequently, several published reports have combined the results of the most highly controlled of these studies and have used meta-analysis, a special type of statistical analysis, to examine these data. A recent report using 34 randomized, controlled trials of exercise-based cardiac rehabilitation programs with >6,000 patients concluded that exercise rehabilitation

significantly reduces total mortality, all-cause mortality, and reinfarction rates among participants.²⁷

The evidence that a comprehensive cardiac rehabilitation program is essential for almost all patients with diagnosed atherosclerotic vascular disease is clear. Evidence-based recommendations state that all eligible patients should be referred to a comprehensive cardiovascular rehabilitation program.²¹ Comprehensive programs should provide a team-based approach to rehabilitation, risk reduction using appropriate behavioral therapies for changing health behaviors and lifestyle, ongoing educational assistance, and technology for tracking patient outcomes and providing feedback and patient support.

In Review

- Deaths during exercise are rare, although typically highly publicized.
 - Deaths during exercise in people over age 35 usually are caused by a cardiac arrhythmia resulting from atherosclerosis.
 - Deaths during exercise in people under age 35 usually are caused by hypertrophic cardiomyopathy, congenital coronary artery anomalies, aortic aneurysm, or myocarditis.
 - Comprehensive cardiac rehabilitation programs facilitate rehabilitation for cardiovascular-related health problems, reduce the risk of subsequent heart attacks, and improve the lifestyle and health behaviors of individuals with cardiovascular disease.
 - These programs should include both aerobic and resistance training components, as well as intensive health behavior change counseling and coaching.
 - The benefits of comprehensive cardiac rehabilitation programs include favorable changes in body composition, glucose metabolism, plasma lipids and lipoproteins, heart function and cardiovascular dynamics, metabolism, health-related quality of life, and reduction in the risk of subsequent heart attacks and death from heart attacks.
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IN CLOSING

In this chapter, we have seen how important physical activity is in reducing the risk for cardiovascular diseases, especially CAD and HTN. We discussed the prevalence of these disorders, the risk factors associated with each, and the ways in which physical activity can help reduce our personal risks. In the next chapter, we continue examining the effects of exercise on health as we turn our attention to obesity and diabetes.

KEY TERMS

adipokines
blood lipids
congenital heart disease
coronary artery disease (CAD)
endothelium
fatty streaks
heart failure
hemorrhagic stroke
high-density lipoprotein cholesterol (HDL-C)
hypertension
ischemia
ischemic stroke
lipoproteins
low-density lipoprotein cholesterol (LDL-C)
myocardial infarction
nitric oxide (NO)
pathophysiology
peripheral vascular disease
plaque
platelet-derived growth factor (PDGF)
primary risk factors
rheumatic heart disease
risk factor
stroke
valvular heart disease

STUDY QUESTIONS

1. What are currently the major causes of death in the United States?

2. What is atherosclerosis, how does it develop, and at what age does it begin?
3. What is hypertension, how does it develop, and at what age does it begin?
4. What is stroke? How does stroke occur? What are the results of stroke?
5. What are the basic risk factors for coronary artery disease? For hypertension?
6. What is the risk of death from coronary artery disease associated with a sedentary lifestyle as compared with an active lifestyle? How has this been established?
7. What are three basic physiological alterations resulting from exercise training that would reduce the risk of death from coronary artery disease?
8. In what ways does endurance exercise training alter risk factors for heart disease?
9. What are the areas covered by the 2013 AHA/ACC practice guidelines with respect to lifestyle management for people with cardiovascular disease?
10. What is a sedentary individual's risk of developing hypertension compared with an active individual's?
11. What are three basic physiological alterations resulting from exercise training that would reduce the risk for developing hypertension?
12. What changes in blood pressure would be expected to result from endurance exercise training in hypertensive individuals?
13. Of what value is cardiac rehabilitation in treating a patient who has had a heart attack?
14. What is the risk of death with endurance exercise training?

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.

